Smartphone-Based WiFi RTT/RSS/PDR/Map Indoor Positioning System Using Particle Filter

Meng Sun, Yunjia Wang, Qianxin Wang, Guoliang Chen, and Zengke Li

***Abstract*— The WiFi fine time measurement (FTM)-based ranging positioning often encounters challenges instability and low accuracy in real-life scenarios due to inaccurate round-trip time (RTT)-based ranging. To address these issues, we construct a WiFi RTT/received signal strength (RSS)/pedestrian dead reckoning (PDR)/map fusion system with an improved map-aided particle filter (PF). Our approach focuses on FTM-based localization from a fingerprinting perspective rather than WiFi ranging positioning. The WiFi FTM fingerprinting utilizes an artificial database, reducing the need for offline labor-intensive measurement work. The semi-parametric error model is employed to compensate for the WiFi RTT and RSS data, and improve fingerprinting accuracy. A collaborative optimization strategy incorporating map and fingerprinting information is designed to enhance the PF's performance in terms of particle impoverishment, propagation, and weight update. Extensive experiments demonstrate that the integrated WiFi RTT/RSS/PDR/map system is error-tolerant under various additional error variance conditions (e.g., heading $\delta_\Psi^2 \leq 4/\text{deg}^2$, or step length $\delta_L^2 \leq 0.5/\text{m}^2$, or RTT ranging $\delta_d^2 \leq 2/\text{m}^2$ and RSS $\delta_R^2 \leq 2/\text{dBm}^2$), with an accuracy fluctuation within 0.1 m. The system obtains meter-level accuracy within 0.19 s when utilizing 1000 particles for every localization request. Compared to the state-of-the-art fusion approaches, the WiFi RTT/RSS/PDR/map fusion system can deliver better accuracy and stability.**

***Index Terms*— Fingerprinting, indoor positioning, map information, particle filter (PF), pedestrian dead reckoning (PDR), sensors fusion, WiFi fine time measurement (FTM).**

I. Introduction

WITH the development of wireless local area networks (WLANs) technologies, WLAN applications are becoming increasingly important in modern society, especially in the indoor location-based services using facilities such as WiFi [1], [2], ultra-wideband [3], Bluetooth low energy [4], and more. As highly integrated devices with various sensors, smartphones are commonly used to deliver consumer-level indoor location-based services. Considering the compatibility and localization investment, performing indoor positioning with WiFi signals on mobile phones is a popular method due to the easy connection and wide-area deployment of WiFi access points (APs). Before the release of the new version of the IEEE 802.11mc WLAN standard [5], classical WiFi positioning was performed by fingerprinting approach [6], which is labor-intensive because a WiFi received signal strength (RSS) database of the interested area should be pre-built for online matching.

Fig. 1 illustrates how the 802.11mc standard enables distance measurement through the utilization of the two-way ranging method. This approach involves measuring the round-trip time (RTT) of WiFi signal flight by using the WiFi fine time measurement (FTM) protocol [7], allowing for the simultaneous capture of distance and RSS data from multiple APs. Currently, several commercial smartphones and WiFi APs have supported the FTM protocol. The primary advantage of WiFi FTM ranging positioning is that it can determine positions with the measured ranging data without the need for a radio map. This convenience reduces the offline measurement work. Therefore, smartphone-based WiFi FTM positioning is becoming more popular in the community.

Researchers have studied various factors influencing ranging such as multipath, sampling burst, bandwidth, and hardware, as discussed in [7]. Additionally, clock deviation, multipath propagation, and non-light-of-sight (NLoS) conditions have been explored in [8]. It has been demonstrated that meter-level WiFi FTM ranging can be reached with appropriate compensation in line-of-sight (LoS) conditions. To improve ranging accuracy, machine learning models like artificial neural networks (ANNs) [9], recurrent neural networks [10], and conventional neural networks (CNNs) are adopted to predict ranging errors and correct ranging data, thereby reducing localization errors. Apart from these, the NLoS/LoS identification is studied to recognize and compensate for or eliminate NLoS errors. Least squares-based support vector machine [11], neural networks [12], and Gaussian model [13] have been introduced to make NLoS/LoS identification and the reported results show that these methods work well regarding improving ranging and positioning accuracy. Although the machine learning-based WiFi ranging compensation and NLoS/LoS identification provide high-precision ranging data for accurate localization, the applications on mobile phones remain limited.

To improve localization accuracy, the integration of WiFi FTM and pedestrian dead reckoning (PDR) is the most popular approach. By leveraging the continuity of PDR, the accuracy

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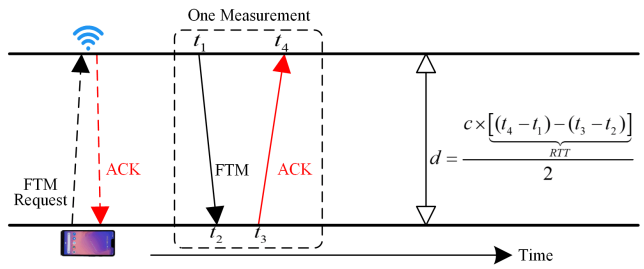

Fig. 1. WiFi FTM ranging process.

and stability of WiFi ranging positioning can be significantly improved [14]. Various filter algorithms, such as the adaptive Kalman filter [15], unscented Kalman filter (UKF) [16], adaptive extended Kalman filter [17], particle filter (PF) [18], [19], have been employed for integrating these two methods. These fusion models focus on optimizing PDR and filter algorithms to solve the low-accuracy problem of WiFi ranging positioning. However, the function of WiFi RSS in localization has not been adequately addressed in these works. Although Guo et al. [20] adopted WiFi RSS into the fusion model, RSS still serves as a supporting role in improving ranging accuracy. Recently, FTM fingerprinting [21] is attracting more attention. The WiFi ranging suffers from complex indoor topologies, but it provides favorable conditions for performing fingerprinting. RTT ranging and RSS data [22], FTM data and corresponding features [21], [23], [24] have been used for fingerprinting purposes. Machine learning algorithms are usually employed in fingerprint matching, such as deep neural networks (DNNs) [21] and support vector machine [25]. However, the machine learning-based fingerprinting approaches have high algorithm complexity and consume more power on mobile phones. Moreover, similar to the shortcomings of traditional fingerprinting methods, constructing a labor-intensive database (or radio map) is still a challenge.

From the aforementioned state-of-the-art approaches, we conclude that ranging compensation, NLoS error identification/elimination, and fusion model construction are the primary research focuses in RTT-based positioning. These efforts aim to overcome the instability of WiFi ranging and positioning caused by complex indoor environments. Although WiFi RSS is simultaneously measured during the ranging process, its role is mainly to support ranging compensation and fusion positioning. In addition to using RSS for RTT ranging compensation, the potential of RSS-based fingerprinting in the FTM localization deserves better analysis. Based on this, WiFi FTM positioning can be achieved not only through the ranging-based approach but also by the fingerprinting method using RSS/RTT. Motivated by these observations, we investigate the following problems in this work.

1) Effective WiFi FTM fingerprinting algorithm and labor-intensive fingerprint measurement work. Developing machine learning models to perform fingerprinting needs lots of WiFi data. How to effectively reduce the offline collection work is significant for the flexibility of fingerprinting applications. Moreover, to deploy the fingerprinting approach on smartphones, it is also necessary to study low-algorithm complexity and efficient fingerprinting algorithms.
2) Particle filter optimization. In practical application, PF often suffers from the degraded filtering problem caused by factors such as particle propagation, particle impoverishment, etc. Improving PFs requires comprehensive consideration of various factors. A high-performance filter model is essential for the fusion of PDR and WiFi positioning.
3) High-precision and robust fusion model construction of WiFi RSS/RTT fingerprinting, PDR, and map data. Based on multisource data and filter model, how to effectively combine these data according to practical applications should be studied. This work should consider various issues such as precision, system complexity, additional error attacks, etc., allowing for stable and efficient operation on mobile phones.

We aim to construct a smartphone-based WiFi RTT/RSS/PDR/map fusion model, which is robust, high-precision, and low-complexity. Different from ranging-based fusion models which mostly rely on PDR methods, we achieve high-precision positioning by optimizing fingerprinting and particle filtering. Based on the above problems, we start with fingerprinting optimization and propose an effective RTT/RSS fingerprinting method without more offline measurement work. We design a collaborative optimization strategy to tackle the degraded filtering problem of the PF. To ensure accuracy and robustness, the map information and PDR are introduced to combine with the RSS/RTT fingerprinting based on the improved PF (IPF) framework. Our contributions are summarized as follows.

1) We propose an effective WiFi RTT/RSS fingerprinting approach, which employs a semi-parametric ranging error model to improve fingerprinting accuracy and eliminate offline labor-intensive measurement campaigns by an artificial WiFi fingerprint database.
2) We propose to improve the PF from the perspectives of particle propagation, weight update, and impoverishment optimization through the comprehensive utilization of map information and WiFi RSS/RTT fingerprinting results at different filtering stages. This optimization strategy allows the PF to maintain effective filtering performance with a low time complexity.
3) We construct a WiFi RSS/RTT/PDR/map fusion system based on the improved particle filter. To the best of our knowledge, we are the first to study the integration of WiFi RTT/RSS fingerprinting, PDR, and map information. This is different from the mainstream research that uses RTT ranging and PDR for fusion positioning.
4) We conduct extensive experiments and observe promising results with the proposed methods. The fingerprinting with artificial radio map achieves an accuracy of about 1.9 m and the fusion localization model delivers a mean positioning accuracy of 0.98 m within 0.19 s. The testing results show that the system reveals tolerance with an accuracy fluctuation within 0.1 m under different error variance conditions (such as heading $\delta_\Psi^2 \leq 4/\text{deg}^2$, or step length $\delta_L^2 \leq 0.5/\text{m}^2$, or RTT ranging $\delta_d^2 \leq 2/\text{m}^2$ and RSS $\delta_R^2 \leq 2/\text{dBm}^2$).

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The remainder of the article is organized as follows: the related works are reviewed in Section II. The system architecture is introduced in Section III. Section IV presents the proposed methods and Section V showcases the testing results. Section VI concludes the work.

II. Related Works

In this section, we focus on WiFi FTM localization and the related fusion methods. Therefore, we review the relevant works of WiFi ranging and positioning, WiFi FTM fingerprinting, and the fusion positioning methods.

A. WiFi FTM Ranging and Localization

The indoor environments affect the WiFi signal propagation, leading to unstable and inaccurate WiFi ranging and subsequently affecting localization accuracy. Ibrahim et al. [7] studied the ranging factors including multipath, sampling burst, bandwidth, and hardware by introducing an open testing platform. Their results demonstrated that meter-level WiFi ranging can be consistently achieved in a low-multipath environment. In [8], clock deviation, multipath propagation, and NLoS factor are further discussed for WiFi FTM ranging, concluding meter-level results in LoS condition. However, indoor complex topologies always lead to serious NLoS and multipath propagation errors on WiFi ranging; this prompts researchers to adopt machine learning models to correct ranging data. In [9], an ANN is developed for RTT ranging and the ANN-based approach outperforms the accuracy obtained by maximum likelihood estimation. In [10], the recurrent neural networks model establishes a network mapping the ranging errors and data, and achieves an average ranging error of 0.5 m. With the simultaneously measured RSS, Guo et al. [26] transformed RSS into distance based on the path loss model (PLM) and combined the transformed distance with RTT ranging data by using a Kalman filter to enhance WiFi ranging accuracy, showing the feasibility of RTT ranging compensation through the fusion of RSS and RTT. Moreover, the regression methods such as least squares (LSs) [14] and nonlinear LSs [27] are studied to compensate for the ranging data.

With the compensated ranging data, trilateration positioning methods including LSs [13] and weighted LSs [14] are used to determine unknown positions. Banin et al. [28] proposed a map-aided WiFi FTM positioning method by using a Bayesian filter, achieving 90% position errors within 3 m with map information. Shao et al. [29] proposed a temporal–spatial constraint method, which transfers a sequence of temporal ranging measurements into multiple virtual positioning clients and solves the positioning estimation with nonconvex optimization, resulting in positioning errors within 1 m in 80% of the cases. To improve positioning accuracy, the NLoS/LoS identification is studied to remove the ranging data with NLoS errors. Dong et al. [11] analyzed RSS and RTT measurement features and tested three machine learning models (LS-SVM, random forest, and neural networks), with neural networks showing higher identification accuracy but requiring more time for model construction. Si et al. [13] used a Gaussian model to distinguish NLoS data, which is removed from the measured dataset. Then, the LSs method is used for positioning and the results show that the positioning accuracy is better after NLoS identification.

These works have demonstrated their advantages in improving ranging positioning accuracy. However, machine learning-based ranging compensation and NLoS classification face limitations in practical applications. The performance of these models depends on real-time measured testing data, which is sensitive to ranging conditions and environments.

B. WiFi FTM Fingerprinting

Due to the influence of indoor topologies, RTT ranging and RSS data obtained from multiple APs at different positions exhibit distinct patterns, allowing for fingerprinting based on WiFi RTT/RSS and their corresponding features. Huilla et al. [30] first investigated FTM fingerprinting using RTT and RSS measurements. They used a weighted k-nearest neighbors method for matching, achieving an accuracy of 2.13 m in 95% of the cases. Bullmann et al. [31] compared the fingerprinting performance by using RTT and RSS, and employed LSs for RTT fingerprinting and probabilistic method for RSS fingerprinting, respectively. Their experiments show that RSS-based positioning obtains better results compared to unoptimized RTT positioning, indicating that RSS can function similar to RTT ranging in delivering positions. Eberechukwu et al. [21] used RTT ranging data and its variance as environment features to make fingerprinting based on a DNN. The DNN model with 600×600 hidden layer size obtains an average positioning accuracy of 0.7 and 0.9 m for the 68th and 95th percentiles, respectively. Park et al. [23] further combined RSS, RTT, and their variances data as input to the DNN model and obtained results of 0.66 and 1.5 m for the 2-σ percentile errors in two datasets. Rizk et al. [24] proposed an *RRLoc* localization system based on the high-level features of WiFi RTT and RSS measurements. The *RRLoc* obtains an average positioning accuracy of 0.51 and 0.59 m in the office and laboratory testbeds, respectively. However, the *RRLoc* system requires more time for location estimation compared to ranging-based methods. Moreover, Martin-Escalona and Zola [25] investigated the performance of RTT and RSS-based fingerprinting by using different classification algorithms (e.g., WKNN and SVM) under different AP layouts, vendors, and frequency bands. Their testing results revealed that the performance of RSS fingerprinting is similar to that of RTT-based positioning. They concluded that complex methods were not necessarily required for fingerprinting, as simpler ones such as WKNN can yield comparable results.

From the aforementioned works, it is evident that machine learning models are popular for FTM fingerprinting. However, algorithm complexity should be considered during practical deployment. Another limitation of these studies is the need for labor-intensive measurement work to build the radio map of the testing area or obtain sufficient data for training learning models.

C. Fusion Localization

The common solution of current fusion localization works is to combine WiFi RTT ranging with inertial measurement units

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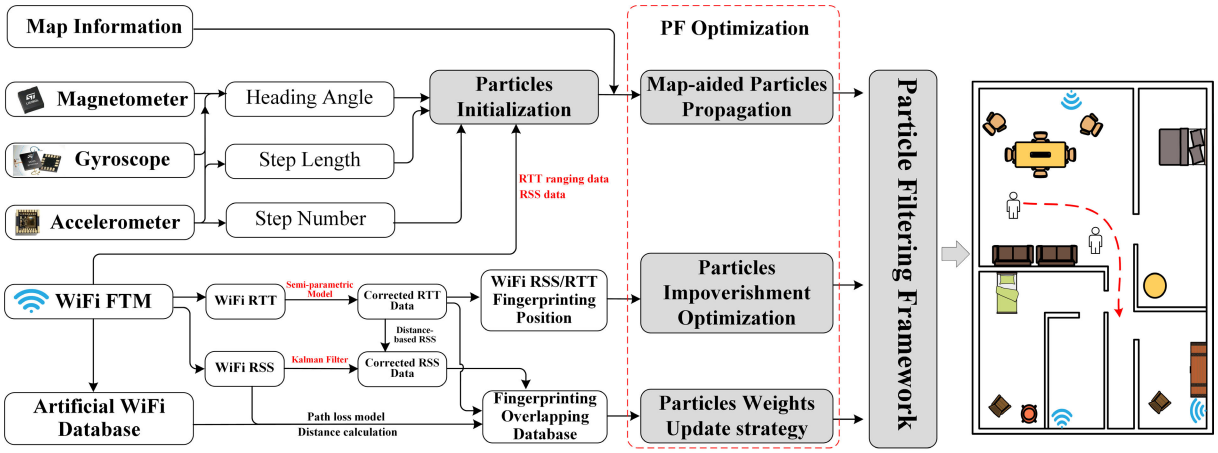

Fig. 2. System architecture of WiFi RTT/RSS/PDR/map fusion.

(IMUs) data using filter algorithms. Liu et al. [15] developed a Kalman filter-based WiFi RTT/PDR fusion model, which incorporates a new outlier detection strategy to improve the RTT positioning accuracy and a real-time smoothing method to assure system robustness, achieving a mean localization error of approximately 1.07 m across four tracks. Yu et al. [16] studied a robust PDR algorithm under different walking states and proposed to combine WiFi FTM ranging positioning with PDR using a UKF. The positioning accuracy improvement of the UKF model relies on the robust PDR. Testing results show that the UKF-based system obtains an accuracy within 2 m in two testing sites. Zhou et al. [17] constructed a WiFi RTT/encoder/INS-based fusion model by using an extended Kalman filter. This model takes advantage of IMU and encoders to maintain high-precision localization; the mean errors are reduced by 54.62% and 58.38% under LoS and NLoS conditions compared to single WiFi RTT ranging positioning, respectively. Similarly, a robust Kalman filter -based framework that combines WiFi RTT, RSS and MEMS-IMU data is proposed in [20], where WiFi RSS is used for WiFi ranging compensation and a positioning evaluation strategy improves robustness, resulting in an average positioning accuracy of 0.572 m.

Besides the Kalman filter models, PF methods are often applied in WiFi RTT/PDR fusion. Yu et al. [18] proposed a sensors fusion framework that uses the unscented PF (UPF) to integrate PDR and WiFi FTM. This sensor model optimizes PDR based on an adaptive extended Kalman filter, which fuses triaxial sensor data to output high-precision PDR information. Testing results reveal that the UPF model obtains a meter-level positioning accuracy in typical indoor scenarios. Xu et al. [19] presented an enhanced PF-based WiFi ranging and IMU sensors fusion method for smartphones. This system addresses the heading estimation optimization for PDR and PF improvement. Testing results indicate that the localization accuracy is within 1 m in 86.7% of the cases. Huang et al. [32] combined a CNN-based WiFi RTT fingerprinting algorithm with smartphone-embedded sensors based on a PF, which is optimized by a map constraint method to update the weights of particles. Although the reported accuracy is better than 1 m in 94.2% of the cases, using CNN and PF will result in high algorithm complexity. In [33], a WiFi/magnetic/PDR fusion system is developed based on the PF. To ensure filtering performance, the initialization of particles is constrained by WiFi RSS fingerprinting, and the particles' weights update is based on magnetic matching. Moreover, WiFi RSS fingerprinting is integrated with PDR by a backtracking PF (BPF) in [34], where the multimodality problem of BPF is solved by using the DBSCAN-based clustering method. The reported accuracy of BPF with clustering in batch mode is 1.6 m. However, the WiFi modules in [33] and [34] cannot provide an RTT ranging function.

In conclusion, the integration of WiFi RTT ranging and PDR is a common solution for developing fusion systems. However, there remains a gap in research regarding fusion using RTT/RSS fingerprinting and PDR. Additionally, optimization for PF performance should be comprehensively considered throughout the filtering process to ensure optimal performance during practical applications.

III. System Architecture

Fig. 2 shows the flowchart of the proposed WiFi RTT/RSS/PDR/map fusion system, which utilizes an IPF. The process begins with WiFi FTM fingerprinting using an artificial database that contains theoretical distance and RSS values between reference points of the positioning area and APs. The construction method is described in Section IV-A. During online matching, the RTT ranging error is compensated using a semi-parametric error model. The compensated RTT data is then used to calculate a distance-based RSS, which is combined with the measured RSS based on the Kalman filter. Pedestrian movement is detected, and step length and heading angles are estimated using a classic PDR framework as in [14]. However, the PDR method does not output location estimation directly in our system, the PDR information (step length and heading angle) is initialized as attributes of particles together with the real-time measured ranging and RSS data.

The PF algorithm embraces WiFi RTT/RSS/PDR/map data to achieve robust and high-precision positioning. Collaborative optimization strategies are designed based on the map information and WiFi FTM fingerprinting results to enhance PF performance in terms of particle propagation, weight update,

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and particle impoverishment. After initialization, the particles propagate using the PDR information, and the map information ensures that particles in inaccessible areas are removed. The weights of particles are updated through RSS and RTT-based fingerprinting using the overlapping database. As the filtering process progresses, particle impoverishment optimization method is employed to replenish the particle swarm. The final position estimation is obtained by averaging the coordinates of the resampled particle swarm.

IV. Methods

A. WiFi FTM Fingerprinting With Artificial Database

1) WiFi FTM Artificial Database Construction: Because the RTT-based distance data and RSS are measured simultaneously, both of them can be used for WiFi FTM fingerprinting. Unlike the labor-intensive collection work of traditional WiFi fingerprinting, we propose to manually construct the database of the target area.

The artificial database consists of theoretical distance and RSS values between the reference points and WiFi APs. During the offline phase, testers should first divide the positioning area into a set of reference points and obtain their corresponding coordinates, denoted as $\{\boldsymbol{G} : g = (x_i, y_i), i = 1, \ldots, t\}$. With n known positions of WiFi APs, the distance between a reference point and an AP can be computed as follows:

$$d_i = \sqrt{\left(x_i - \hat{x}_j\right)^2 + \left(y_i - \hat{y}_j\right)^2} \tag{1}$$

where (x_i, y_i) and $(\hat{x}_j, \hat{y}_j)$ are the positions of the ith reference point and the jth WiFi AP, respectively. A reference RTT fingerprint associated with the ith reference point is represented as a vector $\boldsymbol{d}_i = [d_{i1}, \ldots, d_{in}]^T$. To obtain WiFi RSS, a WiFi PLM should be trained. The relationship between the distance and the WiFi RSS can be modeled as follows:

$$P = P_0 - 10\beta \log_{10}(d/d_0) + \kappa \tag{2}$$

where P and P_0 are the RSS values at the distance d and d_0, d_0 is typically set to 1 m; β is the path loss exponent; and κ is the random error term, respectively. As shown in Fig. 3, due to factors such as hardware, signal propagation, etc., although the WiFi RSS values gradually decrease as the distance increases, different APs have different RSS distributions. The RSS distribution in Fig. 3(c) is more discrete than that in Fig. 3(a) and (b), which means that the PLMs corresponding to different APs should be trained.

To obtain the models, we first measure WiFi RSS values at several ground-truth points during the offline stage, and analyze the RSS data from different APs. P_0 is set to -45 dBm, which is the RSS value when d_0 is 1 m. With the measured RSS values, ground-truth distance, and P_0, the path loss exponent β can be calculated. Based on this process, we trained the PLMs of all the WiFi APs rather than using the same parameters for all APs so that the RSS prediction is more accurate. Based on PLMs, an RSS fingerprint associated with the ith reference point is a vector $\boldsymbol{r}_i = [r_{i1}, \ldots, r_{in}]^T$.

According to (1) and (2), the theoretical distance and RSS values can be calculated. The fingerprint $\boldsymbol{\Xi}_i$ at the ith reference point is expressed as follows:

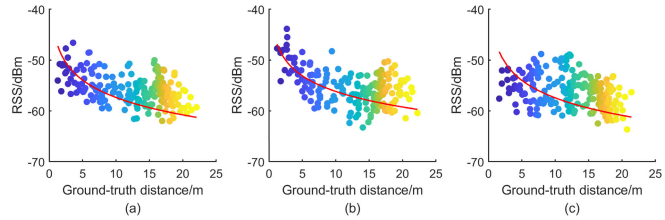

Fig. 3. RSS distributions of three WiFi APs. (a) No. 1 AP, (b) No. 2 AP, and (c) No. 3 AP.

$$\boldsymbol{\Xi}_i = [\boldsymbol{r}_i, \boldsymbol{d}_i]^T = \left[r_{i1}, r_{i2}, \ldots, r_{in}, d_{i1}, d_{i2}, \ldots, d_{in}, x_i, y_i\right]^T \tag{3}$$

where n is the number of WiFi APs, and (x_i, y_i) represents the position of the ith reference point. By performing the same process for all the reference points, we can finally construct the WiFi database of the positioning area without heavy labor-intensive work.

B. Semi-Parametric Error Model-Based FTM Fingerprinting Optimization

The FTM fingerprinting optimization strategy compensates the WiFi data based on a semi-parametric error model [35]. The optimization process first compensates the ranging data by using the semi-parametric model, and then a distance-based RSS value is calculated using the compensated ranging data and (2). Finally, the real-time measured RSS and distance-based RSS are smoothed using the Kalman filter.

In our previous work [35], we have concluded that the WiFi ranging error conforms to the semi-parametric characteristic, which reveals that the ranging error can be generally described as a linear trend related to the distance but with a nonlinear random character. If there are n WiFi APs, the error model is defined as follows:

$$\boldsymbol{E}_{n\times 1} = \boldsymbol{B}_{n\times 2}\boldsymbol{x}_{2\times 1} + \boldsymbol{M}_{n\times 1} + \Delta \tag{4}$$

where $\boldsymbol{E}$ represents the compensated ranging error matrix of n WiFi APs, $\boldsymbol{x}$ denotes the parameter matrix that is further calculated by (6), $\boldsymbol{B}$ is a non-singular matrix that consists of the measured ranging data, $\boldsymbol{Bx}$ describes the linear component; $\boldsymbol{M}$ represents the nonlinear term, and Δ is the measurement noise, respectively. By defining the error equation of (4) as follows:

$$\boldsymbol{V}_{n\times 1} = \boldsymbol{B}_{n\times 2}\hat{\boldsymbol{x}}_{2\times 1} + \boldsymbol{M}_{n\times 1} - \boldsymbol{E}_{n\times 1} \tag{5}$$

where $\hat{\boldsymbol{x}}$ is the matrix of the optimal linear parameters, which is calculated as follows:

$$\hat{\boldsymbol{x}} = \boldsymbol{N}^{-1}\boldsymbol{B}^T\boldsymbol{PE} - \boldsymbol{N}^{-1}\boldsymbol{B}^T\boldsymbol{PM} \tag{6}$$

where $\boldsymbol{B}^T\boldsymbol{PB} = \boldsymbol{N}$, $\boldsymbol{P}$ is a positive-definite matrix. Finally, the nonlinear term $\boldsymbol{M}$ is defined as follows:

$$\boldsymbol{M} = \left(\boldsymbol{P} + \alpha\boldsymbol{R} - \boldsymbol{PBN}^{-1}\boldsymbol{B}^T\boldsymbol{P}\right) - \left(\boldsymbol{P} - \boldsymbol{PBN}^{-1}\boldsymbol{B}^T\boldsymbol{P}\right)\boldsymbol{E} \tag{7}$$

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where α is a scale factor, which is set to 0.01 in this work. More details about the relationships and derivation process of (4)–(7) and the ranging errors prediction process refer to [35].

With the predicted ranging errors and measured distance data, the compensation work is made as follows:

$$\hat{\boldsymbol{d}} = \boldsymbol{d} - \boldsymbol{E}_h - \boldsymbol{E} \tag{8}$$

where $\hat{\boldsymbol{d}}$ and $\boldsymbol{d}$ are the corrected and measured ranging data, both of them contain ranging data from n APs; $\boldsymbol{E}_h$ is the matrix of the APs' hardware errors, which are obtained by calculating the difference between the measured data and the ground-truth distance (e.g., 1 m) in an open area; and $\boldsymbol{E}$ is calculated by using (4), respectively.

For the processing of the real-time measured RSS data, we first make ranging error compensation to obtain the corrected ranging data. Based on (2), a distance-based RSS can be calculated. We employ the Kalman filter to combine the two RSS values as follows:

$$\begin{cases} \boldsymbol{S}_k = \boldsymbol{A}\boldsymbol{S}_{k-1} + \boldsymbol{W}_{k-1} \\ \boldsymbol{Z}_k = \boldsymbol{H}\boldsymbol{G}_{k-1} + \boldsymbol{V}_k \end{cases} \tag{9}$$

where $\boldsymbol{S}_k$ and $\boldsymbol{S}_{k-1}$ are the distance-based RSS values at the time k and $k-1$; $\boldsymbol{G}_{k-1}$ and $\boldsymbol{Z}_k$ represent the measured and filtered RSS; $\boldsymbol{A}$ and $\boldsymbol{H}$ are the transaction and observation matrices, $\boldsymbol{W}_{k-1}$ and $\boldsymbol{V}_k$ are the system and measurement noise, respectively. The RSS filtering of an AP at time k is completed as follows.

1) System state prediction using distance-based RSS

$$\boldsymbol{S}_k^- = \boldsymbol{A}\boldsymbol{S}_{k-1} + \boldsymbol{W}_{k-1}. \tag{10}$$

2) System variance matrix prediction

$$\boldsymbol{P}_k^- = \boldsymbol{A}\boldsymbol{P}_{k-1}\boldsymbol{A}^T + \boldsymbol{Q}_{k-1} \tag{11}$$

where $\boldsymbol{Q}_{k-1}$ is the variance matrix of $\boldsymbol{W}_{k-1}$.

3) System observation estimation using the measured RSS

$$\boldsymbol{Z}_k = \boldsymbol{H}\boldsymbol{G}_k + \boldsymbol{V}_k. \tag{12}$$

4) Compute Kalman gain

$$\boldsymbol{K}_k = \boldsymbol{P}_{k-1}^- \boldsymbol{H}_T \left[\boldsymbol{H}\boldsymbol{P}_{k-1}\boldsymbol{H}^T + \boldsymbol{R}_k\right]. \tag{13}$$

5) System state update using (a), (c), and (d)

$$\boldsymbol{S}_k = \boldsymbol{S}_k^- + \boldsymbol{K}_k\left[\boldsymbol{Z}_k - \boldsymbol{H}_k\boldsymbol{S}_k^-\right]. \tag{14}$$

6) System variance matrix update

$$\boldsymbol{P}_k = (\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H})\boldsymbol{P}_k^- \tag{15}$$

where $\boldsymbol{A}$, $\boldsymbol{H}$, and $\boldsymbol{I}$ are initialized as 1-D identity matrices for the filtering process.

Regarding online positioning, the corrected ranging data and filtered RSS value are combined and evaluated against the artificial WiFi database. The coordinates of the most similar fingerprints are considered as potential position estimations. We employ a k-nearest neighbors (KNN) algorithm [36] to select k similar fingerprints and average their coordinates as the final result.

C. IMU-Based PDR

In our fusion localization framework, the PDR does not work as an independent approach to output positions. Instead, PDR information (e.g., step length and heading angles) is used for the initialization of the particles in the PF. Therefore, the PDR framework in [14] is employed to estimate heading angles, step length, and detect walking steps. These components are essential for the initialization and propagation of particles in our PF-based localization system.

1) Heading Angle Estimation: The heading angle estimation is based on the Mahony complementary filter (MCF) method [37], which is a sensor fusion framework for high-precision walking directions estimation. A quaternion vector $\boldsymbol{q} = [q_0, q_1, q_2, q_3]^T$ is used to represent the attitude of the carrier and the values of $\boldsymbol{q}$ are updated based on the gyroscope data as follows:

$$\begin{cases} q_0(t+T) = q_0(t) - \dfrac{T}{2}\left[\omega_x q_1(t) + \omega_y q_2(t) + \omega_z q_3(t)\right] \\ q_1(t+T) = q_1(t) + \dfrac{T}{2}\left[\omega_x q_0(t) + \omega_z q_2(t) - \omega_y q_3(t)\right] \\ q_2(t+T) = q_2(t) + \dfrac{T}{2}\left[\omega_y q_0(t) - \omega_z q_1(t) + \omega_x q_3(t)\right] \\ q_3(t+T) = q_3(t) + \dfrac{T}{2}\left[\omega_z q_0(t) + \omega_y q_1(t) - \omega_z q_2(t)\right] \end{cases} \tag{16}$$

where T is the sampling time interval, and $q_i(t)$ and $q_i(t+T)$ are the quaternion values at time t and $(t+T)$, respectively. ω_x, ω_y, and ω_z are calculated as follows:

$$\boldsymbol{\omega} = \boldsymbol{\omega}_g + K_p\boldsymbol{e} + K_I\int \boldsymbol{e} \tag{17}$$

$$\boldsymbol{e} = \boldsymbol{e}_a + \boldsymbol{e}_m \tag{18}$$

where $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$ and $\boldsymbol{\omega}_g = [\omega_{gx}, \omega_{gy}, \omega_{gz}]^T$ are the corrected and normalized raw gyroscope data; K_p and K_I represent the error control terms, which are 2.0 and 0.001 in this work, respectively; $\boldsymbol{e}$ is the error compensation of gyroscope data, which is defined by the error terms calculated by accelerometer and magnetometer in (18). Based on the updated $\boldsymbol{q}$ and (16), the heading angle is output as follows:

$$\Psi = tg^{-1}\frac{2(q_1q_2 + q_0q_3)}{1 - 2\left(q_2^2 + q_3^2\right)}, \quad \Psi \in [0, 2\pi). \tag{19}$$

Since the high sampling rates of sensors, we use the mean value of the estimated headings during the one-step period as the walking direction. Our test has shown that the MCF can estimate heading with an accuracy of better than 10° when a pedestrian walks forward by holding a mobile phone horizontally.

2) Step Detection and Step Length Estimation: The step detection in our used PDR framework is performed based on the finite state machine (FSM) algorithm [14], which detects the peaks and valleys of the accelerometer data and is more robust than a peak detection method. The step length is estimated by the well-known nonlinear model [14], [37], as $L = k \times \sqrt[4]{a_{\max} - a_{\min}}$, where k is the scale factor with a value of 0.486 in this work, and $a_{\max}$ and $a_{\min}$ are the peak and valley of the accelerometer data of one step, respectively.

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D. PF Optimization

PF often encounters issues such as filtering divergence, unreliable particle swarm, and particle impoverishment, leading to poor filtering performance and computing resource waste. We propose different strategies to address these problems as follows.

1) Map-Aided Particle Propagation: A PF operates using a particle swarm, where each sample is estimated using a series of previous observations [19]. Every particle is a possible solution for the filtering problem. We randomly assign Gaussian errors to the estimated moving distance and heading angle to allow particles to search in all directions of the positioning area, avoiding error accumulation in one single direction. Therefore, we initialize the kth particle's attributes with the moving distance, heading angle, WiFi RSS, RTT ranging data, and initial weight as follows:

$$\boldsymbol{P}_k = \left\{\tilde{L}_k, \tilde{\Psi}_k, \boldsymbol{r}_{ki}, \boldsymbol{d}_{ki}, w_k, x_k, y_k\right\} \tag{20}$$

where $\boldsymbol{r}_{ki}$ and $\boldsymbol{d}_{ki}$ are the real-time measured RSS and RTT data at the time i, w_k is initial weight, (x_k, y_k) is the particle's position which is randomly selected around the testing area, and $\tilde{L}_k$ and $\tilde{\Psi}$ describe the step length and heading angle assigned with Gaussian errors, calculated as follows:

$$\begin{cases} \tilde{L}_k = L_k + \epsilon_L, & \tilde{\Psi}_k = \Psi_k + \epsilon_\Psi \quad \forall r > 0.5 \\ \tilde{L}_k = L_k - \epsilon_L, & \tilde{\Psi}_k = \Psi_k - \epsilon_\Psi \quad \forall r < 0.5 \end{cases} \tag{21}$$

where r is a random number, $r \in (0, 1)$; $\epsilon_L \sim \mathcal{N}(0, \delta_L^2)$; and $\epsilon_\Psi \sim \mathcal{N}(0, \delta_\Psi^2)$, respectively. When the pedestrian taking steps is detected, the particles move according to

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ L_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cos\tilde{\Psi}_k \\ 0 & 1 & \sin\tilde{\Psi}_k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_k \\ y_k \\ \tilde{L}_k \end{bmatrix} + \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \Delta \end{bmatrix} \tag{22}$$

where σ_1, σ_2, and Δ are the system errors in the x-direction, y-direction, and step length increment, respectively. After one-step propagation, PF evaluates the geometric relationship between the particles' positions and indoor topologies based on the map information. To determine whether to remove a particle outside the map, we introduce a coordinate constraint strategy as follows:

$$\begin{cases} x_k > x_{p\max} || x_k < x_{p\min} \\ y_k > y_{p\max} || y_k < y_{p\min} \end{cases} \quad \forall (x_k, y_k) \in \mathbb{R}^2 \tag{23}$$

where $(x_{p\min}, y_{p\min})$ and $(x_{p\max}, y_{p\max})$ represent the maximal and minimal coordinates of the map boundary, respectively. The advantage of map-aided particle propagation is that it can make the particle swarm propagate within the positioning area and avoid the possible occurrence of divergence.

2) Particle Weights Update With Fingerprinting Based on the Overlapping Database: The particle swarm is evaluated after propagation, ensuring that the PF maintains a highly credible swarm for the subsequent filtering. This evaluation involves assessing the similarities between the attributes of samples in the swarm and the real-time measured WiFi RSS and RTT ranging data. To avoid incorrect evaluation caused by fingerprint mismatching, we propose to perform fingerprinting based on the overlapping fingerprint database to update the weights of particles. This strategy has the advantage of including the coordinates of the samples in the assessment process, which helps obtain more accurate particles close to the ground-truth position. Another advantage of using overlapping fingerprinting data is that the time complexity of assessment work is reduced because the size of overlapping fingerprint database is smaller compared to the entire database or the subdatabase defined by RTT or RSS fingerprinting position.

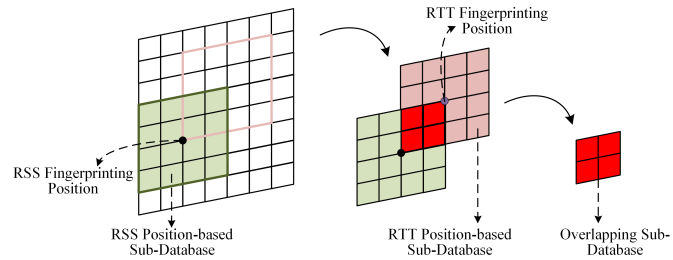

Fig. 4. Overlapping fingerprint database selection.

The detailed process is defined as follows. As Fig. 4 shows, by performing RSS-based and RTT-based fingerprinting, two positions and two subdatabases around these two positions are obtained. The overlapping data is output by selecting the same points from the two subdatabases. It should be noted that if no points are selected after executing this approach, we utilize the RTT position-based subdatabase as the overlapping data. To update every particle's weight, the fingerprint with the closest spatial distance to each particle needs to be first selected from the overlapping database.

As mentioned in Section IV-A, we know that every piece of fingerprint contains WiFi RSS, RTT, and coordinates; the weight of each particle is updated by the following equation:

$$w_i = \frac{1}{\sqrt{2\pi\epsilon}} \exp\left(-\frac{h^2}{2\epsilon^2}\right) \tag{24}$$

where ϵ is the system variance, and h is further given by the following equation:

$$h = \sum_{j=1}^{n} |r_{ij} - R_{ij}| + \sum_{j=1}^{n} |d_{ij} - D_{ij}| + |p_i - P_i|^2 \tag{25}$$

where n is the number of WiFi APs, r_{ij}, d_{ij}, and p_i are the real-time measured WiFi RSS, RTT data, and the position of the ith particle, respectively. R_{ij}, D_{ij}, and P_i are the attributes of the fingerprint with the closest spatial distance to the ith particle. The detailed process of weight update is illustrated in Algorithm 1.

3) Particle Impoverishment Optimization: Particle impoverishment significantly affects the PF's performance and is an inevitable problem because fewer useful particles can be used after multiple resampling. This results in reduced diversity of the particle swarm, which in turn affects the filter's reliability. Thus, it is essential to maintain the diversity and reliability of the particle swarm. To this end, we propose to optimize this problem based on RSS/RTT fingerprinting. To implement this optimization, an evaluation approach should be first executed to determine whether the number of available particles N_{eff}

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Algorithm 1 Particle Weight Update With Fingerprinting Based on the Overlapping Database

Input: particle swarm $pset$, overlapping fingerprint database sub_{fp}, path loss exponent $\boldsymbol{\beta}$, etc.
Output: particle swarm with updated weights $pset'$

```
//Initialization;
Initialize system variance ϵ;
//Execution;
for i ← 1 to length(pset) do
    //select fingerprint;
    for j ← 1 to length(sub_fp) do
        dist(j) ← norm(pset(i).coordinate − sub_fp(j).coordinate)
    end
    index ← find(dist == min(dist));
    //weight update;
    Compensate d̃_i of pset(i) using (4)~(8);
    Calculate distance-based RSS R̃_i using (2) and d̃_i;
    Compensate r̃_i of pset(i) using (9)~(15), and R̃_i;
    Combine compensated d̃_i and r̃_i;
    Find the fingerprint with the minimum distance
    fp{R_i, D_i, P_i} ← sub_fp(index);
    Calculate weight w_i using (24), (25), ϵ, fp;
    Update weight pset'(i).weight ← w_i;
end
```

falls below a predefined threshold N_{thr} after resampling. If this condition is satisfied, the particle swarm will be reinitialized around the RSS/RTT fingerprinting-based position, and the kth particle is reinitialized as follows:

$$\hat{\boldsymbol{P}}_k = \left\{\hat{L}_k, \hat{\Psi}_k, \hat{\boldsymbol{r}}_{ki}, \hat{\boldsymbol{d}}_{ki}, \hat{w}_k, \hat{x}_k, \hat{y}_k\right\} \tag{26}$$

where $\hat{L}_k$, $\hat{\Psi}_k$, $\hat{\boldsymbol{r}}_{ki}$, $\hat{\boldsymbol{d}}_{ki}$, and $\hat{w}_k$ are assigned as the methods in (20) and (21), N_{thr} is the threshold number with a value of 50 in this work, and $\hat{x}_k$ and $\hat{y}_k$ are given by generating random positions around the RSS/RTT fingerprinting-based position, respectively.

This optimization strategy helps to prevent the particle swarm from degenerating into a small subset of particles and ensures the representation of a wide range of possible states. The advantage is that the diversity and reliability are evaluated online, enabling the particle swarm to be refilled and updated promptly to keep effective filtering.

E. WiFi RTT/RSS/PDR/Map Integration Based on the Optimized PF

To construct a high-precision positioning model, we utilize the optimized PF to integrate the RTT/RSS-based fingerprinting, IMU-based PDR, and map information. The detailed fusion process is as follows.

Step 1 (Initialization): The fusion model first loads the map information, RSS/RTT fingerprint database and generates a particle swarm with N samples to uniformly distribute in the localization area. Every particle is initialized as (20). The initial weight is set as $(1/N)$.

It should be noted that we take the RTT/RSS fingerprinting position as the first estimated position of fusion system. When the moving state is detected, the initialization process is finished because PDR information is input, and then the fusion system starts to work.

Step 2 (Map-Aided Particle Propagation): When a pedestrian moves, the particles propagate with the assigned step length and heading angle based on (22). As stated in Section IV-D1, the fusion model will remove the particles outside the map boundaries. The remained particles are passed to the next step.

Step 3 (Particle Weight Update): After propagation, the fusion model uses the approaches in Section IV-D2 to update the weights of particles. This step aims to assign new weights to the particles based on the real-time measured WiFi data. The closer the particle is to the ground-truth position, the higher the weight will be. Finally, the weights should be normalized if all the weights are updated.

Step 4 (Resampling): This step selects the candidates that are possible to be the final estimated position. Generally, a particle with a larger weight will have a higher possibility of being selected. To keep the number of particles consistent, particles with larger weights should be copied multiple times to fill in the particle swarm. The multinomial resampling strategy [38] is employed for particle selection. This operation will result in a loss of particle diversity because samples with larger weights finally occupy the entire particle swarm after multiple copies. Particle impoverishment optimization will address this problem.

Step 5 (Position Estimation): The fusion position is output by computing the average coordinates of the resampled particles

$$\begin{cases} X_k = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} x_k \\ Y_k = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} y_k \end{cases} \tag{27}$$

where n denotes the number of resampled particles.

Step 6 (Particle Impoverishment Optimization): From Step 4, the diversity of the swarm will be reduced with the filtering going because resampling copies samples lead to a loss of particle diversity. Therefore, after the position estimation, it is necessary to assess the number of available samples to optimize the impoverishment problem. As stated in Section IV-D3, a new particle swarm will be regenerated if the kinds of available particles are fewer than a threshold.

For the sequential fusion positioning process, the system executes Steps 2~6 to integrate WiFi RTT/RSS/IMU/map data. This enables simultaneous position estimation while optimizing the PF. The detailed process is illustrated in Algorithm 2.

V. Experiments and Results

A. Experimental Setup

Fig. 5 shows that the testing experiments were conducted on the second floor of a typical real-life environment, with the targeted area measuring 24.75 × 27.4 m. Eight WiFi RTT

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Algorithm 2 WiFi RTT/RSS/PDR/Map Fusion

Input: particle number N, real-time measured RSS $\hat{\boldsymbol{r}}_k$, RTT data $\hat{\boldsymbol{d}}_k$, Step length scale factor k, error control terms of heading estimation K_p and K_I, map data, fingerprint database, path loss exponent $\boldsymbol{\beta}$, Kalman filter parameters $\boldsymbol{A}$, $\boldsymbol{W}$, $\boldsymbol{H}$, $\boldsymbol{V}$, threshold number N_{thr}, etc.

Output: fusion positioning result (X, Y)

```
//Initialization;
Load map information and fingerprint database;
Initialize N particles within the testing area;
//Fusion;
while new measurements received do
    Compensate d̂_k according to (4)~(8);
    Compensate r̂_k using corrected d̂_k, (2), and KalmanFilter;
    Calculate heading Ψ and step length L;
    Encode particles with (20);
    Performing fingerprinting and obtaining positions;
    if new step then
        Particles propagation based on (22), (23) and map data;
        Particles' weights update using Algorithm 1;
        Multinomial Resampling;
        Calculate available particles N_eff;
    end
    if N_eff > N_thr then
        Calculate fusion result using (27);
    else
        Particle impoverishment optimization;
        Regenerate particles using (26);
        Particles' weights update using Algorithm 1;
        Multinomial Resampling;
        Calculate fusion result using (27);
    end
end
```

APs (type: Compulab fitlet2) are installed outside of the floor with a height of 5.2 m from the ground, resulting in an NLoS condition. The height of the second floor is 0.56 m higher than that of the WiFi AP. Therefore, to estimate 2-D location, it is necessary to transform the measured ranging data considering the heights of the floor and APs. We selected 10 ground-truth points and measured the accurate coordinates by using a total station. An interpolation method is used to obtain 1756 grid points (GPs) with a separation of 0.3 m. To train a PLM, we measured WiFi RSS data at 172 GPs. With the trained models, the coordinates of GPs and WiFi APs, we constructed an artificial WiFi RTT/RSS fingerprint database by using the method in Section IV-A. The testing points are selected every 2.4 m and the corresponding ground-truth coordinates are measured by using the total station.

A Google Pixel 3 phone that supports WiFi FTM is used for testing. The sampling rates of WiFi RTT ranging and IMU sensors are 5 and 50 Hz, respectively. The number of particles

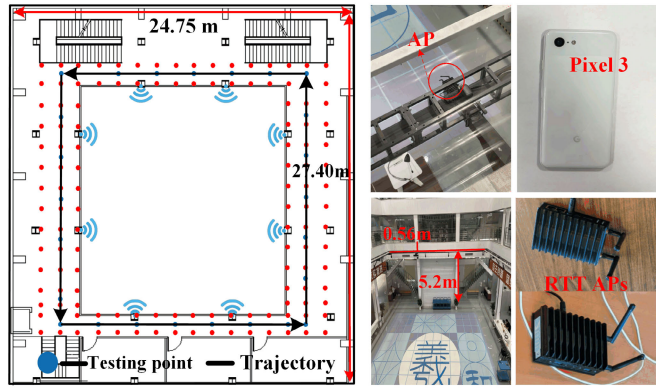

Fig. 5. Floor plan and real-life scenario of the experimental area and the testing device.

TABLE I
POSITIONING ERRORS OF FINGERPRINTING METHODS

Compensation	Methods	Mean/(m)	RMSE/(m)	90th/(m)
Semi-parametric	RSS	2.59	3.12	5.31
	RTT	2.50	2.94	4.48
	RSS/RTT	1.91	2.46	4.00
/	RSS	2.92	3.36	5.34
	RTT	2.60	3.05	4.54
	RSS/RTT	2.23	2.70	4.41

for PF is 1000. Regarding the size of the sub-fingerprint database for particle weight updating (Section IV-D), we set 2 m as the searching radius to define the subdatabases. Tester walked along the reference path for three cycles and recorded the WiFi and IMU data. The testing data is analyzed on a laptop with 8 GB RAM and 2.6 GHz CPU.

B. Fingerprinting Performance Evaluation

We first investigate the proposed fingerprinting approach against the fingerprinting method without error compensation. The testing results are arranged in Table I and Fig. 6. We utilize the meaning positioning error, root-mean-square error (RMSE), and 90th percentile error to evaluate the performance.

Table I reveals that directly performing fingerprinting using WiFi RSS and RTT achieves an accuracy within 3 m at our testing site. The accuracy of the mixed RSS/RTT fingerprinting is 2.23 m, which is improved to 1.91 m with compensation based on the semi-parametric model. RSS-based fingerprinting has the most improvement of 0.33 m. With error compensation, the RMSEs of all fingerprinting approaches are reduced, which means that the stability is improved. The semi-parametric model also reduces the 90th percentile errors of these three fingerprinting methods, especially with the mixed RSS/RTT fingerprinting, which obtains a reduction of 0.41 m. Fig. 6 also shows that mean errors and standard errors of positioning results are mitigated after error compensation. The testing results prove the beneficial impact of semi-parametric model-based data compensation on improving fingerprinting accuracy.

To further investigate the localization results of ranging-based and fingerprinting methods, we compare the proposed solution with other approaches such as WKNN-based fingerprinting [30], as well as ranging-based methods

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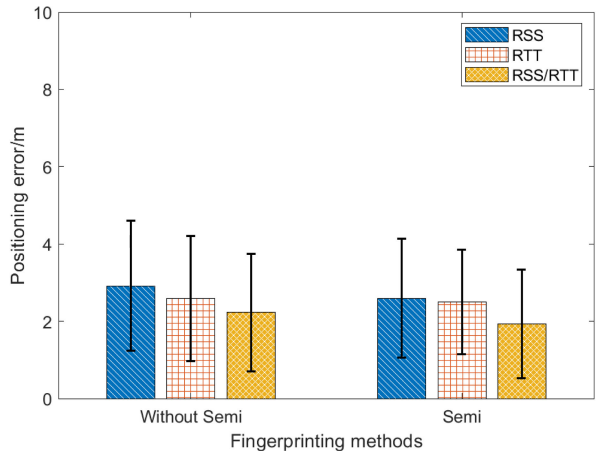

Fig. 6. Fingerprinting comparison with/without compensation.

TABLE II
POSITIONING ERRORS OF RANGING AND FINGERPRINTING METHODS

Methods	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
Proposed	1.91	2.46	2.48	4.00
WKNN [30]	2.03	2.56	2.68	4.18
RSS-FTM [26]	3.25	3.74	4.17	5.00
LS [16]	3.77	4.37	5.09	6.59

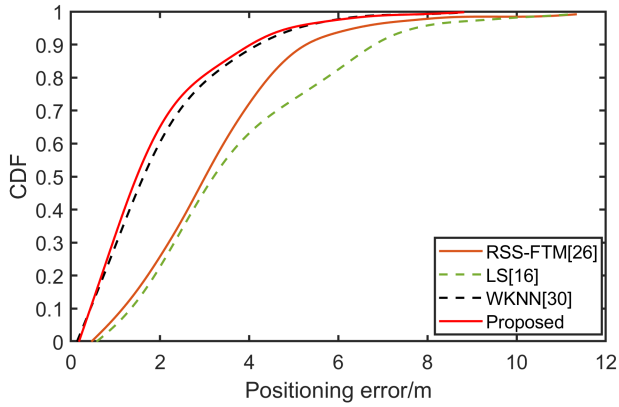

Fig. 7. CDFs comparison of ranging and fingerprinting methods.

including RSS-FTM fusion [26] and LSs [16]. The matching data used is the mixed RTT/RSS. The results are presented in Table II and Fig. 7.

Table II shows that fingerprinting outperforms ranging-based methods, highlighting the advantages of fingerprinting in real-life NLoS scenarios. Both the proposed method and WKNN achieve an accuracy of approximately 2 m, whereas the ranging-based methods have positioning errors of over 3 m. In terms of RMSE, 75th and 90th percentile errors, fingerprinting also surpasses ranging-based methods. Fig. 7 describes the superiority of FTM fingerprinting compared to the ranging-based approaches. We analyze that it is because the fingerprint database is built within the testing area and the matching results are expected to fall within this area. However, ranging-based methods are not limited by this constraint, resulting in large positioning errors due to uncorrected ranging data. Furthermore, in comparison to the WKNN method, our method offers better positioning performance without the need for labor-intensive measurement work, thus providing more flexibility in building and updating the fingerprint database.

TABLE III
POSITIONING ERRORS OF THE FUSION SYSTEM

Times	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
1	1.03	1.20	1.32	1.93
2	0.90	1.07	1.19	1.70
3	1.01	1.16	1.29	1.80
4	0.91	1.08	1.12	1.52
5	1.02	1.18	1.26	1.90
6	0.93	1.06	1.18	1.59
7	1.03	1.23	1.29	1.91
8	0.96	1.10	1.28	1.71
9	1.03	1.21	1.33	1.85
10	1.00	1.18	1.36	1.86
Average	0.98	1.07	1.13	1.49

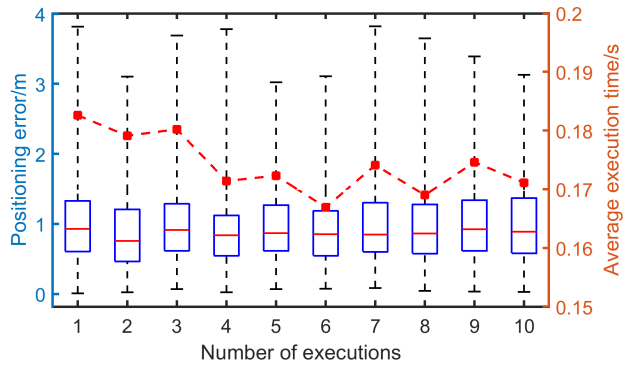

Fig. 8. Fusion positioning results and AET of every localization request.

C. Fusion Localization Performance Evaluation

In this section, we evaluate the positioning accuracy and time complexity of the fusion system through 10 localization runs. Table III and Fig. 8 present the positioning results and average execution time (AET) of the fusion system. The average results are obtained by calculating the mean values from the 10 sets of positioning results. The variances for the initialization of particles in terms of heading and step length are $\delta_{\Psi}^2 = 1^\circ$ and $\delta_L^2 = 0.1$.

Table III shows that the proposed system achieves a mean positioning error ranging from 0.9 to 1.03 m across the 10 localization runs, resulting in an average error of 0.98 m. The RMSE ranges from 1.06 to 1.23 m, with a mean RMSE of 1.07 m. The mean 75th and 90th percentile errors reveal that the proposed system can deliver localization results with errors within 1.13 and 1.49 m at confidence levels of 75% and 90%, respectively. The testing results validate the good performance of the proposed fusion framework.

Fig. 8 displays error boxes representing the uniform distribution of the 10 sets of positioning results without significant fluctuations. The red lines within the error boxes indicate that the proposed system achieves a median localization accuracy of around 1 m. The largest error among the 10 sets of positioning results remains within 4 m. The red dashed line represents the AET of each localization request, fluctuating between 0.16 and 0.19 s. This indicates that the proposed fusion framework exhibits low time complexity, enabling long-time usage on smartphones. In summary, the integration

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TABLE IV
POSITIONING ERRORS OF THE SYSTEM WITH PDR ERROR ATTACKS

$\delta_\Psi^2/(deg^2)$	$\delta_L^2/(m^2)$	Mean/(m)	RMSE/(m)	90th/(m)
$\mathcal{N}(0,1)$	$\mathcal{N}(0,0.1)$	0.98	1.07	1.49
$\mathcal{N}(0,4)$	$\mathcal{N}(0,0.1)$	1.06	1.16	1.74
$\mathcal{N}(0,8)$		1.16	1.32	1.99
$\mathcal{N}(0,16)$		1.22	1.39	2.08
$\mathcal{N}(0,30)$		1.26	1.44	2.17
$\mathcal{N}(0,1)$	$\mathcal{N}(0,0.2)$	1.01	1.10	1.61
	$\mathcal{N}(0,0.3)$	1.04	1.14	1.66
	$\mathcal{N}(0,0.4)$	1.05	1.14	1.63
	$\mathcal{N}(0,0.5)$	1.07	1.17	1.66

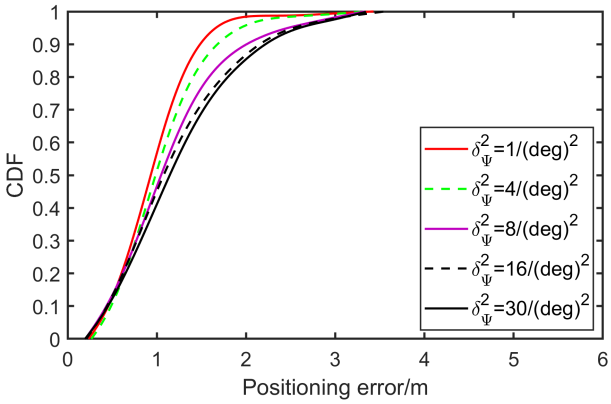

Fig. 9. Fusion positioning results with different heading biases.

of WiFi RTT/RSS fingerprinting with IMU sensors and map information proves to be feasible and efficient. The fusion model demonstrates low time complexity while achieving meter-level positioning accuracy.

D. Fusion Localization Performance With PDR Error Attacks

Based on (21) and the original output PDR information, we utilized different δ_Ψ^2 and δ_L^2 to simulate PDR error attacks and assess the robustness of our system. The localization method is executed 10 times under every condition as shown in Table IV. Then, we calculate the mean results of the 10 sets of positioning results for comparison.

Table IV presents that the mean localization accuracy is 1.06, 1.16, 1.22, and 1.26 m when the original output heading with the variances of 4, 8, 16, and 30 with units of [/deg^2], showcasing a negative correlation between system accuracy and heading errors. The RMSEs also increase as the heading errors increase, reaching 1.44 m when $\delta_\Psi^2 = 30/\text{deg}^2$. Similarly, Fig. 9 illustrates the same trend for the 50th to 90th percentile error when comparing different heading conditions. We also observe that the mean positioning accuracy and RMSE are reduced by 0.08 m (8.16%) and 0.09 m (8.41%) when $\delta_\Psi^2 = 4/\text{deg}^2$, demonstrating relative stability against the condition $\delta_\Psi^2 = 1/\text{deg}^2$. Considering the presence of original headings errors, we conclude that although introducing additional heading bias affects positioning performance, our system maintains robustness within a certain range of heading variance (e.g., $4/\text{deg}^2$).

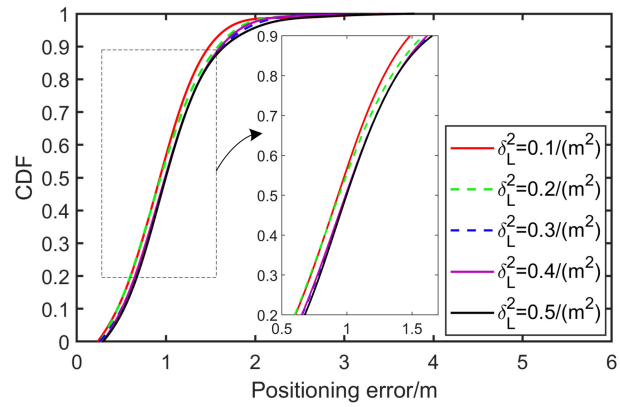

Fig. 10. Fusion positioning results with different step length biases.

TABLE V
POSITIONING ERRORS OF THE SYSTEM WITH WIFI ERRORS

$\delta_d^2/(m^2)$	$\delta_R^2/(dBm^2)$	Mean/(m)	RMSE/(m)	90th/(m)
/	/	0.98	1.07	1.49
$\mathcal{N}(0,0.5)$	$\mathcal{N}(0,0.5)$	1.01	1.11	1.61
$\mathcal{N}(0,1)$	$\mathcal{N}(0,1)$	1.03	1.12	1.51
$\mathcal{N}(0,2)$	$\mathcal{N}(0,2)$	1.07	1.18	1.61
$\mathcal{N}(0,3)$	$\mathcal{N}(0,3)$	1.13	1.21	1.57

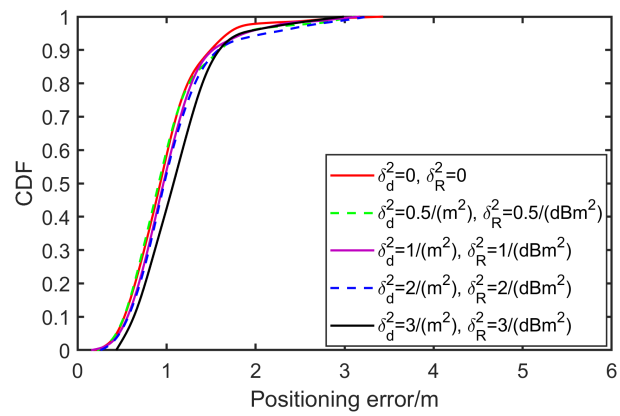

Fig. 11. Fusion positioning results with different WiFi biases.

When adding difference variances (0.2, 0.3, 0.4, and 0.5 with units of [/m^2]) to the original estimated step lengths, Table IV shows a reduction in mean localization accuracy of 0.03, 0.06, 0.07, and 0.09 m for the proposed system, respectively. The largest RMSE difference is 0.1 m (about 9.34%). These results demonstrate that the proposed system is error-tolerant to step length errors. Fig. 10 reveals similar cdf curves of different δ_L^2 conditions, affirming the robustness of the fusion system when setting different step length error attacks.

E. Fusion Localization Performance With Additional WiFi Data Errors

To assess the fusion system's performance under various WiFi observation error scenarios, we introduced additional Gaussian errors to the system. We utilize a random number r to control whether errors should be added. If r is greater than

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TABLE VI
DESCRIPTIONS AND CONFIGURATIONS OF THE COMPARED STATE-OF-THE-ART METHODS

Methods	Descriptions	Configurations in this work
EKF-based RTT/PDR [14]	A tightly coupled WiFi RTT/PDR fusion model using an EKF. An adaptive WiFi RTT positioning method is used to improve RTT positioning accuracy.	Coefficients for ranging model: -0.058, -0.1334, -0.0012. EKF parameters: $\boldsymbol{A} = 0.75 * [1, 0, -sin(\theta); 0, 1, cos(\theta); 0, 0, 1]$, $\boldsymbol{P} = diag\{1, 1, 1\}, \boldsymbol{B} = diag\{0.25, 0.25, 0.25\}$, $\boldsymbol{Q} = diag\{0.05, 0.05, 0.05\}$, $\boldsymbol{R} = diag\{1, 1, 1, 1, 1, 1, 1\}$
EPF-based RTT/PDR [19]	WiFi RTT and PDR are integrated using an enhanced particle filter, which is optimized in terms of particle weight update with map constraints and WiFi RTT outlier detection.	The parameters for the step length model: $a = 0.371$, $b = 0.227$, $c = 1$. The threshold for the peak-detection-based step detection is $10.8m/s^2$. The parameters for RTT ranging outlier detection: $d_s = 1.2m$, $\eta = 1$. The number of particles is 1000.
EKF-based Magnetic/PDR [37]	An EKF-based magnetic/PDR fusion system. The magnetic positioning is optimized by an enhanced mind evolutionary algorithm and PDR is improved in terms of gait recognition using an extreme learning machine.	EMEA parameters: the size of the population is 100. The numbers of superior subgroups and temporary subgroups are 2 and 8. EKF parameters: $\boldsymbol{A} = [1, 0, -sin(\theta); 0, 1, cos(\theta); 0, 0, 1]$, $\boldsymbol{P} = diag\{1, 1, 1\}, \boldsymbol{B} = diag\{1, 1, 1\}$, $\boldsymbol{Q} = diag\{0.1, 0.1, 0.1\}$, $\boldsymbol{R} = diag\{0.1, 0.1, 0.1\}$.
IPF-based RSS/Magnetic/PDR [33]	The magnetic positioning and PDR are combined using an improved particle filter (IPF). WiFi RSS fingerprinting is first executed to assign the initial position of the system and define the iterative window (IW) and constraint window (CW), which are used for improving PF.	The sizes of IW and CW are 2.4 m× 2.4 m and 9.6 m ×9.6 m, respectively. The number of particles is 1000. The value of K for KNN-based RSS fingerprinting is 3. The step length is fixed at 0.6 m and the threshold for the peak-detection-based step detection is $10.8m/s^2$.
KF-based RSS/Magnetic/PDR [39]	WiFi RSS and PDR are combined using the Kalman filter. Magnetic data is used for corner detection and a trajectory optimization method is proposed to further optimize fusion results.	The threshold for corner detection is 50°. The contained steps for trajectory optimization are 6. KF parameters: $\boldsymbol{A} = [1, 0, cos(\theta); 0, 1, sin(\theta); 0, 0, 1]$, $\boldsymbol{P} = diag\{1, 1, 1\}, \boldsymbol{B} = diag\{1, 1, 1\}$, $\boldsymbol{Q} = diag\{0.05, 0.05, 0.05\}$, $\boldsymbol{R} = diag\{1, 1, 1\}$.

Note: $\boldsymbol{A}$ and $\boldsymbol{B}$ are the transition and measurement matrices, $\boldsymbol{P}$ and $\boldsymbol{Q}$ denote the matrices of system variance and noise, $\boldsymbol{R}$ is the variance matrix of the measurement noise, respectively. For details of these approaches, please see the corresponding references.

TABLE VII
POSITIONING ERRORS OF DIFFERENT METHODS

Methods	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
PDR	2.23	2.42	3.04	3.42
Proposed	0.98	1.07	1.13	1.49
EKF-based RTT/PDR [14]	1.27	1.59	1.62	2.51
EPF-based RTT/PDR [19]	1.18	1.35	1.55	2.10
IPF-based RSS/Mag/PDR [33]	1.29	1.61	1.52	2.01
EKF-based Mag/PDR [37]	1.03	1.22	1.40	2.05
KF-based RSS/Mag/PDR [39]	1.15	1.28	1.50	2.06

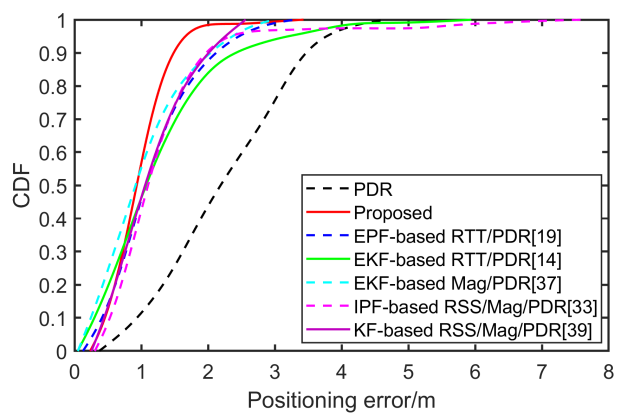

Fig. 12. CDFs comparison of different methods.

0.5 ($r > 0.5$ and $r \in [0, 1]$), one or up to seven WiFi APs are selected, and errors with different variances are simultaneously added to RTT and RSS data from the selected APs. The system is executed 10 times under different error conditions, and the results are summarized in Table V and Fig. 11.

As illustrated in Table V, the mean accuracy decreases as the variances of RTT and RSS data increase, reaching 1.13 m when $\delta_d^2 = 3/\text{m}^2$ and $\delta_R^2 = 3/\text{dBm}^2$. The fluctuation of the mean positioning error is within 0.1 m when the variances of RTT and RSS data are less than 2 m and 2 dBm, respectively. Fig. 11 also presents similar cdf curves of the positioning results when subjected to the condition of $\delta_d^2 \leq 2/\text{m}^2$ and $\delta_R^2 \leq 2/\text{dBm}^2$. The results reflect that WiFi data errors caused by scenarios reduce the positioning performance of the proposed system. However, with the evaluation method that the mean accuracy fluctuation is no more than 0.1 m, the system can still be considered error-tolerant under additional WiFi biases with variances of $\delta_d^2 \leq 2/\text{m}^2$ and $\delta_R^2 \leq 2/\text{dBm}^2$.

F. Comparison With State-of-the-Art Approaches

To further analyze the performance of the proposed system against state-of-the-art methods, we implemented WiFi RTT/PDR fusion systems using EKF [14] and an enhanced PF (EPF) [19], as well as an advanced WiFi RSS/magnetic/PDR fusion system [33] using an IPF, an EKF-based magnetic/PDR fusion system [37], and the WiFi RSS/magnetic/PDR fusion method [39] using a Kalman filter (KF). It should be noted that the WiFi RSS fingerprinting is performed using an artificial database as stated in Section IV-A. Each fusion system is executed 10 times and the positioning results are averaged. Table VI presents descriptions and configurations of

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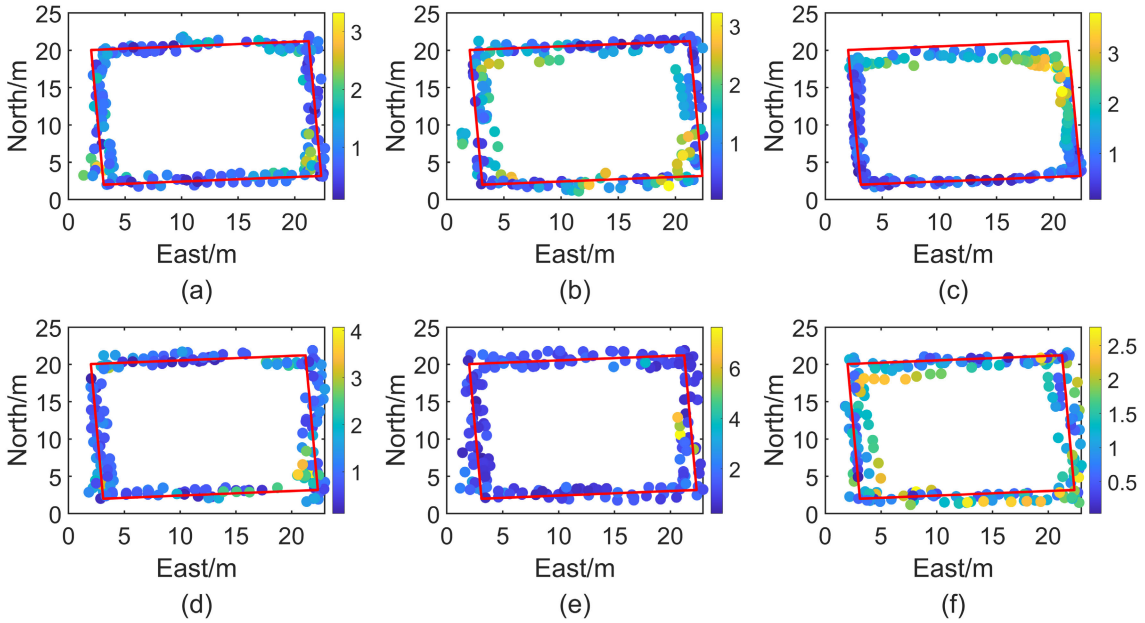

Fig. 13. Errors of the output positions of different fusion systems. (a) Proposed system, (b) EKF-based Mag/PDR, (c) EKF-based RTT/PDR, (d) EPF-based RTT/PDR, (e) IPF-based RSS/Mag/PDR, and (f) KF-based RSS/Mag/PDR. The red line is the ground truth, and the color points represent the position errors with units of meter.

the compared systems. Table VII summarizes the positioning errors, and Fig. 12 compares the CDFs.

The integration of WiFi and magnetic data significantly improves PDR positioning accuracy. For the comparison of WiFi RTT/PDR fusion systems, the mean localization error and RMSE of the EPF-based system [19] are 1.18 and 1.35 m, respectively, which are 0.09 and 0.24 m lower than those of the EKF-based system [14]. This difference is evident in the CDFs shown in Fig. 12, indicating larger localization errors in the EKF-based RTT/PDR system.

In the comparison of RSS/magnetic/PDR fusion systems, the KF-based integration [39] achieves an accuracy and RMSE of 1.15 and 1.28 m, which are 0.14 and 0.33 m better than those of the IPF-based fusion [33]. The KF-based RSS/magnetic/PDR fusion also achieves better 75th and 90th percentile errors, which can be confirmed in Fig. 12. This improvement is attributed to the IPF-based system relying solely on magnetic fingerprinting for particle weight updates after using RSS fingerprinting for the initial system position (see Table VI). However, a KNN method and 3-D magnetic features may not ensure accurate magnetic matching, thereby affecting the PF filtering.

Although another EKF-based magnetic/PDR framework obtains an average accuracy of 1.03 m, our system has smaller RMSE, 75th, and 90th percentile errors. Moreover, our system also outperforms the compared five state-of-the-art fusion methods regarding mean accuracy, RMSE, 75th, and 90th percentile errors. Fig. 12 illustrates that the proposed approach achieves superior results from the 60th to 90th percentile errors compared to the available state-of-the-art methods. These results demonstrate that integrating FTM fingerprinting (through RTT and RSS), PDR, and map information can achieve better positioning accuracy. Therefore, high-precision WiFi FTM/PDR fusion can be implemented not only by integrating WiFi RTT and PDR but also by combining FTM fingerprinting with PDR.

G. Discussion

Although WiFi FTM ranging provides a new solution for indoor location-based services, its accuracy is often compromised in complex indoor environments. In our testing site, the 90th percentile positioning error of the RSS-FTM fusion method in [26] is only 5 m, which indicates that the instability of WiFi ranging cannot ensure high-precision WiFi positioning. The complex indoor environments also limit the practical application of the NLoS/LoS identification method [11] on mobile phones. Based on our testing results, FTM fingerprinting shows better localization performance compared to the ranging-based approach, which means that FTM fingerprinting can work independently as the RTT-based ranging method. The artificial database used in this work also allows for better flexibility during practical application.

We prove that integrating WiFi RSS/RTT fingerprinting with IMU data is an efficient smartphone-based positioning approach that can estimate locations with meter-level accuracy. Instead of focusing on improving localization accuracy through WiFi ranging and PDR optimizations as seen in [14] and [19], the proposed system optimizes the particle filter with map and FTM fingerprinting, and finally obtains meter-level accuracy and better stability. Fig. 13(a) illustrates that the proposed system produces more stable positioning results, with fewer large errors occurring around the ground-truth trajectory compared to the systems in [14], [19], [33], [37], and [39]. This stability is further confirmed by the RMSE values in Table VII, which reveals that the proposed system achieves the lowest RMSE among the six fusion approaches. In contrast to approaches that aim to improve system performance

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through PDR optimization in [16] and [18], our collaborative optimization strategy for PF proves to be effective during testing. When faced with additional PDR and WiFi errors (see Sections V-D and V-E), the system demonstrates tolerance within an accuracy fluctuation of 0.1 m when subjected to error variance conditions such as heading $\delta_\Psi^2 \leq 4/\text{deg}^2$, or step length $\delta_L^2 \leq 0.5/\text{m}^2$, or RTT ranging $\delta_d^2 \leq 2/\text{m}^2$ and RSS $\delta_R^2 \leq 2/\text{dBm}^2$. Moreover, the AET of the proposed PF-based system is less than 0.19 s, indicating its suitability for long-term deployment on mobile phones. It can be summarized that the proposed WiFi RTT/RSS/PDR/map fusion model offers low time complexity and robustness, and can obtain meter-level accuracy in real-life scenarios.

VI. Conclusion

In this work, we proposed a PF-based sensor fusion system based on WiFi RSS/RTT/PDR/map data for smartphones and proved that performing WiFi FTM fingerprinting can also obtain considerable localization results in indoor complex environments. The proposed fusion system delivers a mean accuracy of 0.98 m, showing better stability compared with the ranging-based fusion methods. Moreover, the fusion calculation is finished within 0.19 s, which means that the model can be deployed on mobile phones for a long time. The future works will go to the PDR optimization to provide more accurate IMU information and explore the possibility of performing the high-precision collaborative fingerprinting and ranging positioning approach, and test the system in more real-life scenarios.

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